

$$3. \quad c(t) = 10 \cos(2\pi f_c t) \quad m(t) = \cos(20\pi t)$$

$$a) \quad u(t) = 10 \cos(2\pi f_c t + k_f \int_0^t m(\tau) d\tau)$$

$$\int_0^t \cos 20\pi \tau d\tau = \left[\frac{\sin 20\pi \tau}{20\pi} \right]_0^t = \frac{\sin 20\pi t}{20\pi}$$

$$\Rightarrow u(t) = 10 \cos(2\pi f_c t + \frac{5}{2\pi} \sin 20\pi t)$$

$$(b) \quad \text{Power of } u(t) : \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T u^2(t) dt = \frac{(10)^2}{2} = 50 \text{ Watts.}$$

$$\text{Deviation ratio : } \frac{\Delta f}{B} = \frac{k_f \max_t |m(t)|}{2\pi(B)} = \frac{k_f \cdot 1}{2\pi(B)} = \frac{50}{2\pi(B)}$$

$$c) \quad \text{Carson's rule: } B_{FM} = 2B + 2\Delta f \quad B = 20 \text{ Hz.}$$

$$= 2B \left(1 + \frac{5}{4\pi} \right)$$

$$4. (a) \quad f_{c\theta} = f_c + f_{IF} = (4 + 850) \text{ to } (6 + 850) = 4.85 \text{ MHz to } 6.85 \text{ MHz}$$

$$(b) \quad f_c + 2f_{IF} : \text{Image freq: } 4 + 2f_{IF}, 6 + 2f_{IF}$$

$$= 4 + 1.700 \text{ MHz}, 7.7 \text{ MHz.}$$

$$= 5.7 \text{ MHz}$$